Bogdan Gavrilov

Machine Learning Engineer / Data Scientist / RecSys Engineer

dill862409@mail.ru @bogdan_kinovari LinkedIn +7 (922) 259-03-08

Overview

Machine Learning Engineer with **3+ years of experience** building **recommendation systems**, **ranking models**, customer analytics and predictive ML solutions. Experienced in designing end-to-end ML pipelines, feature engineering, **retrieval and ranking architectures**, and deploying data-driven products in retail and biomedical domains.

Skills Summary

- **Programming Languages**: Python, SQL
- **RecSys & Ranking**: Personalization, Ranking, Retrieval, Candidate Generation, Re-Ranking, Collaborative Filtering, Matrix Factorization, Sequential Recommendations, Two-Tower Models, Learning-to-Rank
- **Machine Learning**: Classic ML, Feature Engineering, A/B Testing, Search, Clustering, Time Series, NLP, Prompt Engineering, Fine-tuning
- **Deep Learning**: PyTorch, TensorFlow, Embeddings, Representation Learning, Metric Learning, Distillation, Quantization
- **Libraries & Frameworks**: Scikit-learn, CatBoost, XGBoost, LightGBM, Pandas, NumPy, SciPy, Polars, implicit, LightFM, RecBole, FAISS, FastAPI, PySpark
- **MLOps & Infrastructure**: Docker, Airflow, MLflow, Weights & Biases, Git, Linux, Monitoring, Jupyter Notebook
- **Databases**: PostgreSQL, MySQL, SQLite, ClickHouse
- **Platforms**: Windows, MacOS, Kaggle, Google Colab
- **Languages**: Russian (Native), English (B2)

Experience

- **Machine Learning Engineer | TechAudit LLC** Moscow, Russia
  08.2024 - Present
  Built recommendation and predictive ML systems for retail clients, including personalization, ranking, churn prediction, and substitute recommendation pipelines.
  - **Personalized Recommendation Systems**: Developed ML-based recommendation pipelines for retail clients (including SPAR), implementing candidate generation and CatBoost ranking for personalized product recommendations, improving **CTR by 8.5%**.
  - **Basket & Cross-Sell Recommendations**: Built recommendation models for shopping cart personalization, engineering user, basket, and product features to improve cross-sell performance, increasing **Average Basket Size by 6.5%**.
  - **Substitute & Similar Products**: Developed item-to-item recommendation models for out-of-stock scenarios, combining retrieval strategies and ranking models to suggest relevant product alternatives, improving **Recovery Rate from 10.8% to 12.6%**.
  - **Customer Churn Prediction**: Built churn prediction models for retail customers using behavioral and transactional data to support targeted retention campaigns, reducing **churn from 31.5% to 29.6%**.
- **Machine Learning Engineer | BIOCAD JSC** Saint Petersburg, Russia
  02.2023 - 07.2024
  Developed machine learning models and analytical pipelines for biomedical research using structured clinical and laboratory data.
  - **Treatment Response Prediction**: Developed machine learning models for predicting patient response to therapy using clinical and laboratory data, improving **ROC-AUC from 0.68 to 0.78** through feature engineering, preprocessing, and model optimization.
  - **Patient Stratification**: Applied dimensionality reduction and clustering techniques (**PCA, K-Means, Hierarchical Clustering**) to identify patient cohorts and support exploratory biomedical research and personalized treatment analysis.
  - **Data Processing & Model Evaluation**: Built preprocessing pipelines for medical datasets, handled missing values, feature extraction, statistical analysis, and model evaluation using classification metrics for internal R&D initiatives.

Education

Emperor Alexander I Saint Petersburg State Transport University (PGUPS) Saint Petersburg, Russia
Specialist in Business Informatics & Data Analysis
GPA: 5.0 / 5.0

Courses

Karpov Courses: Hard ML	2025
MTS: Your Second RecSys	2024
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Deep Learning School (Semester 1 — Classic ML & CV)	2021
Karpov Courses: Python Fundamentals	2021